

ME44305 System Analysis and Engineering

Assignment 1 & 2

Carbon Neutral Festivals - Group 1

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1 Background and literature

The UK Music Industry Greenhouse Gas Emissions (2008) estimates the total greenhouse gas emissions of the UK music industry to be at least 540,000 tonnes of CO₂ in 2007. It is estimated that 43% of these emissions are caused by visitor travel to festivals, arenas and venues [1]. The first part of the research concludes that the music industry should put more priority towards climate responsibility in this part of the music industry. However, it is difficult to put full responsibility toward the music industry as travel choices are made by the visitors, thus the emissions are only indirectly in the control of the industry. We will research and model the possibilities around including public transport prices in the ticket price for the festival itself.

A researcher working with several festivals in the UK has also taken a look into this [1]. For a festival in the UK, several visitors were asked what their primary reason was to travel by coach (public transport). The results were as follows. Most coach goers travel by coach only because a combined coach and festival ticket was the only ticket type available (72.1% $\pm$ 3.9%). Convenience was a significant factor (44.9%), while very few people chose to travel by coach for reasons of cost (14%) (n=626). Other findings were that the service flexibility differed per festival, overall the coach travelers were sufficiently happy with their coach service (good-very good), coach travelers are likely to travel by coach again, and the communication found on the festival website was important to motivate visitors to travel by coach.

2 Innovative Idea

Our innovative idea which we are developing in cooperation with Ribs and Blues Raalte moves further in this idea. When buying festival tickets online, visitors are asked for their intended mode of transport to the festival. The options are public transport & car. Visitors who will travel by public transport will later receive a discount code that can be used to purchase their train ticket. With this discount code, visitors can buy a return ticket from any NS station (NS stands for Dutch Railway) in the Netherlands to NS station Raalte. The return ticket must include Raalte station as the destination to ensure that the discount code cannot be used for non-festival related travel. Evidently, visitors who will travel by car will not receive a discount code. Those arriving by car must additionally buy a car park ticket.

An additional fee is added to the original festival ticket price for all visitors. The revenue generated from these fees is used to finance the discount on public transport tickets for visitors traveling by train. A research conducted by the Dutch Association of Dutch Pop Music Venues and Festivals has shown that it is essential to integrate travel information in the ticket process, because that is when most visitors determine their mode of transport [2]. Therefore, during checkout, it is clearly shown that there is an additional fee which is separate from the original festival ticket price. The fee is justified by explaining that the combined contributions are intended to stimulate the use of public transport among festival visitors. Because all visitors pay the fee (regardless of their chosen mode of transport), together we will reward those who travel by public transport and penalize car use. The additional fee is shown and explained separately to increase awareness among visitors about their mode of transport.

The Dutch Kennisinstituut voor Mobiliteitsbeleid (KiM) has examined the price sensitivity of public transport demand in response to fare increases [3]. In this study, the institute reconstructed historical developments to isolate the specific effect of price changes on public transport usage from other factors influencing demand. The results indicate that a 2.8% increase in train fares for travelers leads to a 1.3% decrease in the number of kilometers traveled by train [3].

Another research carried out by KiM examines the elasticities of the price of domestic passenger mobility [4]. Important conclusions of the study indicate that a 10% reduction in train ticket prices leads to a 4.4% increase in train use and a 0.7% decrease in bus, tram and metro (BTM) use. Furthermore, the findings suggest that travellers generally respond more strongly to travel time incentives than to price incentives. In particular, an increase in car travel times due to congestion appears to be a much stronger stimulus for travellers to switch to public transport than changes in ticket prices. These findings suggest that travel time incentives may be particularly relevant to explore further, and could be valuable to consider when developing or refining the model.

Kennisinstituut voor Mobiliteitsbeleid (2015) (Knowledge Institute for Policy around Mobility) carried out a literature study on sensitivities in public transport demand that shows when travel time increases by 10%, public transport use decreases by approximately 4–7%, depending on the transport mode and context. The study also demonstrates that travelers perceive time spent outside the vehicle as more burdensome than time spent inside the train, metro, or bus. Consequently, factors such as walking time, waiting time, transfer time, and the unreliability of public transport schedules weigh relatively heavily in the perceived attractiveness of public transport[5].

In this study, visitor behavior of visitors of Ribs and Blues Festival Raalte, who are traveling to the festival, will be modeled by adjusting parameters such as transport prices and arrival times. These parameters will be used to simulate visitor flows and estimate crowd sizes at key locations, such as train stations, ticket desks, parking areas, and other access points.

3 System definition

The system is the transportation network that moves festival visitors from their homes to the Ribs and Blues Festival in Raalte. It includes the ticket sale process, where the festival can influence travel choices, the resulting mode split, physical infrastructure (roads, stations, parking and shuttle busses), and crowd flows at key access points along the routes (train station, bus stops, parking, and festival access). The system can be modeled using the following eight key components: the visitors, the visitor generator, the road network, the parking lot, the train station, the shuttle bus, the walking routes, and the ticket scan. The system boundary starts at the moment a visitor purchases a ticket online and ends the moment they pass through the festival entrance. Underneath a short description of each component is written.

3.1 CATWOE analysis

The CATWOE analysis will help us understand the system from the point of view of different stakeholders. It stands for Customers, Actors, Transformation, Worldview, Owners, and Environment.

The **customers** of the system are the festival visitors for whom the festival is organized. They are the main source of income for the festival. They are affected by travel costs, travel lengths, and overall system experience.

The **actors** are the ones who execute operations within the system. These are the festival organization (Ribs and Blues), the NS and the shuttle bus operators, as these parties carry out the activities that make the system function.

The **transformation** in the system encourages visitors to choose more sustainable transportation to the festival, by making public transport more attractive. This will consequently reduce CO_2 emissions and help the festival towards carbon neutrality.

The **world view** is that festivals have a collective responsibility to reduce their climate impact.

The **Owner** of the system is the festival organization (Ribs and Blues), as they carry the responsibility and ownership of the festival. They distribute tickets, take initiative in the organization, and are the contact point for visitors/customers. They have the authority to change or even cancel the system (festival). To implement our innovation, the festival organizer will have to agree on our innovative idea's.

The **environment** consists of the constraints within our system must operate, including the physical infrastructure such as NS station Raalte or the local road networks. The constraints of our system are described in Section 3.3.

3.2 Components description

1. Visitor

The main entity is a visitor, which represents an individual festival visitor traveling from their origin to the festival. Visitors have their mode of transport (car & public transport) and can receive a discount code (train) or a parking ticket (car). Here, we try to make predictions about human behavior. Decision variables that influence visitor transport choices are the: festival ticket price, mandatory sustainable fee, sustainable awareness of visitors, train ticket discounts, car travel costs, parking ticket cost, expected travel time for car & public transport, and convenience factors such as waiting, transfer, and reliability.

2. Road network

The road network represents the local road infrastructure between roads leading into Raalte and the parking lot.

3. Ns station

The NS station Raalte receives visitors traveling by public transport.

4. Shuttle bus

The shuttle bus transports visitors from the Raalte NS station to the festival terrain.

5. Walking routes

The walking routes represent the connections from the drop-off points to a next key access point. For example: NS station Raalte $\rightarrow$ Shuttle bus $\rightarrow$ Ticket scan, or parking $\rightarrow$ ticket scan. Each arrow represents a walking route. It depends on the layout of Raalte and the festival terrain.

3.3 Constraints

The transportation system operates within a set of constraints that bound what the system can achieve and how it behaves. These constraints arise from the physical infrastructure available in and around Raalte, the behavior of the festival visitors, and the financial feasibility to support the discount mechanism.

1. Infrastructure

NS Raalte is a small station with limited platforms and shuttle bus capacity. Similarly, the local road network and parking lot have a maximum throughput after which congestion and queuing becomes severe. These capacities should not be exceeded.

2. Financial

The sustainable Fee must be such that it will fund the train discounts without making tickets unaffordable.

3. Behavioral

Travelers respond more strongly to travel time than to price (KiM, 2024), so price incentives alone have limited effect.

4. Legal/practical

Discount codes must be restricted to NS routes that end in Raalte to prevent misuse.

3.4 Performance Metrics

To evaluate the system, we define a set of performance metrics. These metrics capture the system's performance across three elements: 1. the environmental impacts, its operational efficiency, and the financial viability. Together, these 3 elements provide a basis for comparing different parameters within the simulation model.

Sustainability performance metrics

- modal split: % of visitors arriving by train vs car
- transport emissions per visitor ($\text{gCO}_2/\text{visitor}$)
- number of cars per visitor

Operational efficiency performance metrics

- Queue lengths at key access points
- waiting times at key access points
- Peak crowd density at bottleneck nodes (station, ticket scan)
- Road congestion level on Raalte's local roads

financial performance metrics

- Revenue from sustainable fees vs cost of train discounts (budget balance)

3.5 Black Box model

The black box model, as shown in figure 1, gives a visual overview of how different input factors within the system are transformed to relevant output indicators. The inputs include ticket and pricing information, behavioral parameters (namely price elasticities), visitor data and characteristics of the transport network and external conditions.

The black box itself represents the system in which these inputs are translated to output indicators. The model outputs include both sustainability indicators, such as CO_2 emissions per visitor and modal split) and operational and system performance indicators.

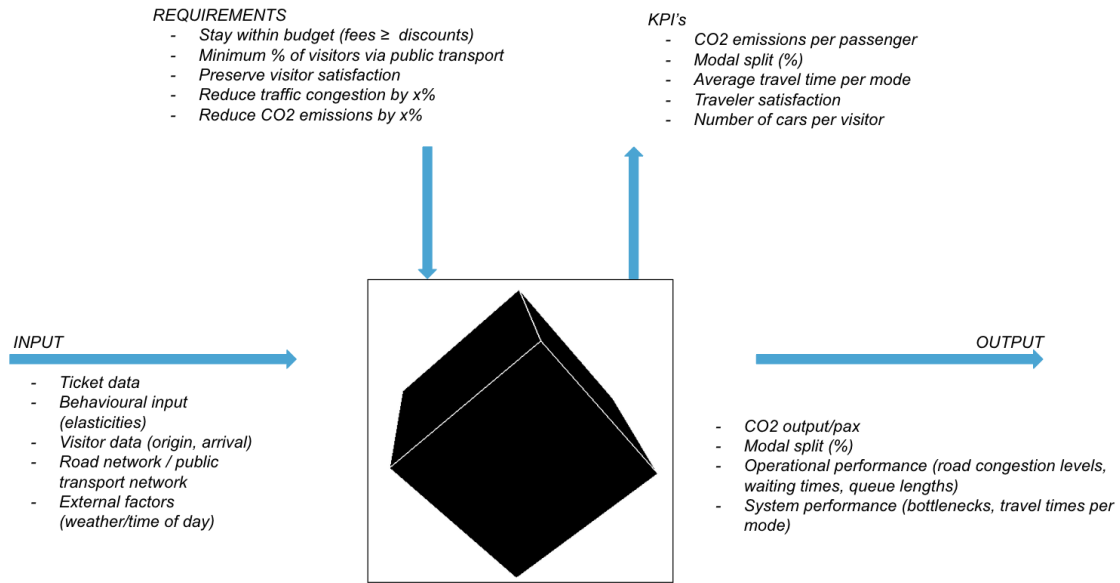


Fig. 1. Black box model

4 Conceptual model

4.1 The Decision Module (Pre-Festival Ticket Phase)

This part of the model determines how people choose to travel, calculating the modal split based on the elasticities derived from the literature review. The simulation consists of two stages. A pre-simulation decision model that determines the modal split, and the dynamic discrete simulation of visitor flows that uses the predicted modal split as input. The decision model is run once before the simulation starts, translating parameters such as discount rates, travel-time estimates, and parking fees into an expected modal split using the earlier mentioned elasticity literature. Higher discount rates and higher parking fees will push more visitors to public transport, for example.

Trigger The process is initiated when a visitor enters the online ticket shop. Visitors decide their transport mode at the moment of ticket purchase or well before actually traveling towards the event (VNPF, 2026). The general transport choices are effectively fixed before the festival starts. This justifies computing the modal split once before the simulation rather than dynamically during it. There are exemptions for visitors who change transport type at the last minute, but this effect is neglected in our research, since we expect this effect to be minimal.

Cost Inputs

- **Base Festival Ticket Price + Mandatory Sustainability Fee:** Applies to all attendees.
- **Car Route Cost:** Fuel cost + Parking Ticket.
- **Train Route Cost:** Standard NS Ticket - Discount Code (Funded by the sustainability fee).

Behavioral Modifiers (Based on KiM literature)

- **Price Elasticity:** A 10% drop in train prices leads to a 4.4% increase in train use and a 0.7% drop in other transit (KiM, 2024). Car users with alternatives are highly sensitive to these price drops (KiM, 2018).
- **Travel Time Elasticity:** Increased car travel time (congestion) is a stronger trigger to switch to OV (public transport) than price (KiM, 2024). A 10% increase in overall travel time decreases OV use by 4–7% (KiM, 2015). This may have a similar inverse effect: a decrease in travel time could increase OV usage.
- **Convenience Penalty:** Out-of-vehicle time (waiting for the shuttle, walking from the station) is perceived as significantly heavier than in-vehicle time (KiM, 2015).

Output of this module A percentage breakdown of the total expected visitors sorted into two primary groups: Car and public transport. This output will be the input of the dynamic discrete model.

4.2 The Dynamic Flow Module (Journey & Arrival)

This module provides a time-based simulation of the roads and infrastructure in Raalte.

Car Flow (Road Network)

- **Nodes:** Highway exits → Raalte local roads → Parking lot entrance → Walk to festival gates.
- **Dynamic Variable:** As more cars enter, local road congestion increases. This increases car travel time, which can theoretically be fed back into the Decision Module for future ticket buyers or real-time route planners.

Train Flow (Public Transport Network)

- **Nodes:** Arrival at NS Station Raalte → Out-of-vehicle transfer → Shuttle bus or Walking route → Festival gates.
- **Dynamic Variable:** Train arrival times create “pulsed” crowds (e.g., a large group arrives every 30 minutes).

Bottleneck Simulation The model calculates queue lengths and waiting times at the parking gates, the shuttle bus stops, and the final ticket scan based on the volume of arriving visitors.

4.3 Model Outputs

The output of the model integrates the data from both systems. By feeding the decision data into the dynamic flow simulation, the following outcomes can be evaluated:

- **Traffic in Raalte:** Useful for monitoring if the overall traffic volume in the city effectively decreases.
- **Crowd Heatmap:** Identifies peak busyness at the parking lot, NS station, and bus station.
- **CO₂ Reduction:** The ultimate goal of the assignment. Calculates the difference in CO₂ emissions achieved when people choose public transport over driving.

5 Conceptual Simulation Model

The Swim-lane model is visualized in Figure 8. The Swim-lane is made up of six lanes:

1. Visitor Generator
2. Visitor
3. Car Generator
4. Car
5. Shuttle Bus
6. Ticket Scan

5.1 Visitor Generator

Events The visitor generator contains a primary arrival event responsible for spawning visitor entities. The generation rate is governed by a time-varying module, where arrival volumes are adjusted at each time step to reflect fluctuating demand.

Queues There is one dedicated queue associated with the generator. This queue handles the creation of visitors based on a stochastic distribution, where the inter-arrival times are sampled to simulate realistic, non-uniform visitor arrivals.

Process: Logic, Sequence, and Dependencies The generation logic is driven by empirical data from the festival. Once generated, the entities are passed directly into the visitor swim lane, initiating their movement through the system. Each generated visitor is assigned a transport mode (car or public transport) based on the model split ratio produced by the decision module described in Section 4.1. In the PDL, this corresponds to the `ModePublicTransport : DetermineSplit`.

5.2 Visitor

Events The primary events for the visitor entity include their arrival at the station or arrival at the Raalte road network, entering/exiting queues, the ticket scan process, and their final departure from the system.

Queues There are two primary queues the visitor interacts with:

The shuttle Bus Queue: Upon arriving at the station platform, visitors who choose public transport proceed to the shuttle bus stop. Because train arrivals are discrete and pulsed (based on the NS timetable), this queue experiences batch arrivals, resulting in sudden surges rather than gradual growth. The waiting time is a direct function of the synchronization between train arrivals and the shuttle bus departure schedule. If a train arrives immediately after a shuttle departs, the entire batch remains in the queue until the next departure.

Ticket Scan Queue: After completing their chosen mode of transport, visitors merge back into the main process and enter a queue for the ticket scan before exiting the system.

Process: Logic, Sequence, and Dependencies **Logic:** The core logic of this module revolves around the visitor's modal choice. The decision to use a car, the shuttle bus, or to walk is handled by a decision module grounded in behavioral literature.

The visitor arrives at the station and the decision module determines their transport mode. If the shuttle bus is chosen, the visitor walks to the stop and enters the shuttle bus queue. (If the car is chosen, the process follows subsection 5.4. The visitor boards the transport and briefly leaves the main visitor swim lane to enter the transport swim lane. Upon completing the journey, all visitors merge back into the main visitor swim lane. The visitor queues for the ticket scan, undergoes the scanning event, and finally departs the system.

Dependencies: This process has two strict dependencies on other swim lanes. First, a visitor can only board a shuttle if a bus entity is physically present at the stop (dependent on the Bus swim lane, discussed in subsection 5.5). Second, the visitor's progression through the final stage is dependent on the availability and processing speed of the ticket scan module.

5.3 Car Generator

Events The car generator has one event, which creates a car. This event creates the car object with an attribute for the amount of passenger space and an attribute for the starting location.

Queues There is one queue in the car generator lane, which is the wait for generating a car. This wait time is based on how long it takes to fill a car, as the system wants to ensure there is always a car waiting. As soon as the previous car leaves its spot, a new car is generated on that spot.

Process: Logic, Sequence and Dependencies This lane is to ensure that there are cars available. The sequence is to wait for an open spot and then generate a car with its attributes in there. It is dependent on whether the car has moved away from its spot in the car lane.

5.4 Car

Events The car has a few events along its path from start to parking lot. The first event fills the car with the visitors in the queue. Then an event places the car at its spawn node based on the cars attribute. Another event is when the car arrives at the parking lot. Finally, the visitors leave the car.

Queues During its travel, the car has a few queues. First it waits for the queue of visitors to be large enough to fill the car. Then, the car goes on the road and travels from node to node at a predetermined route. When the parking lot is reached, the car queues at the entrance. After that, the car queues to find a space in the parking lot. The car lane is dependent on the visitors that are generated and the car that is generated and in the end the lane returns the visitors back to their lane.

Process: Logic, Sequence and Dependencies The car is used to move the visitors from their starting place to the parking lot. This is done by first getting a number of passengers in a car to simulate carpooling. Then, the car is being spawned at its starting location based on its attribute. The car travels over the road network from node to node until it arrives at the parking lot. Then it has to queue to enter the parking lot and find a parking space. After the car is parked, the visitors exit the car and the car stays in its spot.

5.5 Shuttle Bus

Events The first event is a scheduled bus departure event, where the shuttle bus arrives at Raalte station and fills up to maximum capacity with visitors from the shuttle bus queue. After the bus is at maximum capacity or the queue is empty, the bus begins its journey from NS Station Raalte toward the festival area near the parking lot. In addition, there are node arrival events along the route. As the bus travels from node to node, it may encounter other traffic and traffic lights where it must queue before it can proceed. When the bus arrives at the parking lot, another event will release all passengers from the bus. Finally, a return event will make the empty bus travel back from the parking lot at NS Station Raalte, passing through the nodes and traffic network again, after which it becomes available for the next departure.

Queues The shuttle bus faces multiple queues. Most of these queues appear along the shuttle bus route itself. At traffic light nodes, the bus must wait before it can continue. The queues affect the bus travel time and therefore the turnaround, which in the evening determines how quickly the bus becomes available again at the station. If the bus gets delayed in traffic, the efficient shuttle frequency decreases and the queue for passengers at the stations grows.

Process: Logic, Sequence and Dependencies The shuttle bus process follows this sequence:

1. The bus waits at the station for passengers until the departure time or until it reaches maximum capacity.
2. The bus departs and travels from node to node along the road network towards the parking area
3. At each node, the bus checks to see whether it has arrived at the parking lot. If it did not arrive yet, it may encounter a traffic light queue where it waits before continuing to the next node.

4. When it arrives at the parking area, all visitors leave the bus and proceed to the "walk to ticket scan" step in the Visitor lane
5. The bus then makes the return trip to the station via the road network again traveling from node to node and queueing at the traffic lights.
6. Once back at the station, the bus waits as an empty bus until it is needed again for the next departure.

The upstream dependency is the NS station and the Visitor lane. Public transport visitors traveling by shuttle bus must arrive at the train station. The demand for shuttle buses is driven by the arrivals of trains. The downstream dependency is the ticket scan. When a full bus unloads at the parking area, it releases a group of visitors who arrive at the ticket scanner simultaneously, creating a surge in capacity demand. An important interaction with the Car lane also exists: the shuttle bus shares the road network and traffic lights with cars. During peak car arrivals, traffic light queues become longer, slowing the bus turnaround and reducing effective shuttle capacity. This means that heavy car traffic indirectly worsens the public transport experience

5.6 Ticket Scan

Events The ticket scanning process has two events. The first is a visitor arrival. The second event is a scan event. When a scanner finishes processing a visitor, the visitor exits the system, and the scanner becomes available to the next person in the queue.

Process: Logic, Sequence and Dependencies The sequence of the ticket scan is described below.

1. A scanner becomes available, the first visitor in the queue is going in the ticket scan process/
2. The ticket process needs time to process the visitor.
3. After the process time has passed, the visitor leaves the ticket scan, and thus leaves the system.
4. The ticket scan becomes available again

The ticket scan is entirely dependent of upstream components as it is the last process in the system. It is the only component in the entire swim-lane model where all transport paths converge, making it the final bottleneck of the system. The Shuttle Bus lane feeds it with groups of visitors at intervals. The Car lane feeds it with a more continuous but variable stream that increases in throughput during peak arrival hours. Because these upstream processes operate independently with their own timing, capacity constraints, and traffic light delays, the ticket scan has to absorb all the output they produce. The number of scanners is therefore a critical design parameter for the experimental plan. Too few and the queue grows uncontrollably. Too many scanners and scanners sit idle between arrivals.

Explain swimlane model Define the processes within each component and describe their logic, sequence, and dependencies. o Explain how different processes interact within the system

6 Experimental Plan

This chapter develops an experimental plan outlining how the simulation will be used to analyze system behavior and evaluate the proposed innovation.

For the experimental plan, the idea is to test different factors to stimulate visitors to choose public transport, instead of choosing the car. These factors are the following:

- Factor 1: Discount on train tickets. Varying from 0 to 100% where the decision is based
- Factor 2: Sustainability fee. Varying from 0 to 40% of the original ticket price.
- Factor 3: Frequency of the shuttle bus. Varying from 3 to 6 times per hour.
- Factor 4: Parking costs. Varying from 0 to 50% of the original ticket price.

By varying these factors across scenarios, the decision model recalculates a new modal split for each combination, and the dynamic simulation is then run with that updates split to analyze the effect on the system and the overall Carbon emissions.

For the sustainability fee and train discount, we will base the behavioral data of potential visitors on the data of the research we found. It is possible to design a model of behavioral choice behind this, but this is beyond the scope of this project.

The experiments aim to answer two questions.

- Does the proposed innovation (sustainability fee + train discount) meaningfully shift modal split toward public transport and reduce CO_2 emissions?
- Which parameter has the strongest effect: the discount size, the sustainability fee, shuttle bus frequency, or parking cost?

To be able to draw conclusions from the questions above, the new situation of the innovative idea needs to be compared to the original situations. That is why a baseline model needs to be made to represent the current situation without the innovation. This means running the model without train ticket discount, sustainability fee, frequency of the shuttle bus and parking costs. This is the baseline scenario where the modal split as observed for Ribs and Blues, which we will gather from the data in assignment 3.

All experimental scenarios are compared against this baseline to measure the effect of changes. These experimental scenarios are tested as different combinations, to see the impact of the differences. These scenarios are: - Financial balance: Compare the baseline against a small set of financially balanced discount-fee combinations. For example: 25% discount funded by a €2.50 fee, 50% discount funded by a €5.00 fee, and so on. This isolates the innovation's effect with roughly 4–5 scenarios. -Sensitivity analysis: For the most promising scenario from Layer 1, vary one factor at a time while holding others constant. This tells which parameter in the system is most sensitive to. It varies the train discount across its full range, then the shuttle frequency across its full range, and so on. - Extremes test: Test extreme but plausible conditions: a sold-out festival with maximum visitor numbers, a train cancellation removing one or two scheduled services, or a shuttle bus breakdown reducing available buses mid-simulation.

7 PDL

```

1 # Global Parameters
2 NumberVisitors:           integer
3 EventEndingTime:          integer
4 RoadNetwork:              list of (NodeID, NextNodeID, Distance)
5 TrafficDensity:           list of (NodeID, CarCount)      # car count on the next
                        section
6 TrafficSpeedFunction:     f(CarCount) -> Speed           # based on traffic-flow
                        model
7 CarFree:                  boolean                        # semaphore: True when
                        a car slot is available
8 MyCarQueue:               Queue                          # shared queue visitors
                        join when travelling by car
9 WalkToShuttlebus:         integer                       # fixed walk time from
                        entry to shuttle stop
10 WalkToTicketScan:        integer                       # fixed walk time from
                        drop-off to ticket scan
11 ParkingLotEntryDelay:    TExponentialDistribution       # delay per car at
                        parking lot entrance
12 FindSpaceParkCar:        integer                       # time per car already
                        parked to find a space
13 BoardingTimePerPassenger: integer                      # time for one
                        passenger to board the bus
14 AlightingTimePerPassenger: integer                    # time for one
                        passenger to alight the bus

```

TVisitorGenerator

```

1     InterDepartTime:      TExponentialDistribution
2     ModePublicTransport:  TDeterminedSplit              # fraction using
                        public transport
3
4 Process
5     Repeat
6         Wait InterDepartTime.sample
7         newVisitor = new TVisitor
8         newVisitor.MPT = ModePublicTransport.sample      # assign transport
                        mode
9         newVisitor.DepartTime = Now
10        newVisitor.Start

```

TVisitor

```

1     MPT:                  boolean      # True = public transport, False = private car
2     DepartTime: integer
3     ArrivalTime: integer
4
5 Process
6     if MPT == True:
7         Wait WalkToShuttlebus
8         myShuttlebus.MyBusQueue.EnterQueue(self)
9         Passivate                      # reactivated by the shuttle-bus process
10    else:
11        MyCarQueue.EnterQueue(self)
12        Passivate                      # reactivated by car process
13    Wait WalkToTicketScan
14    myTicketScan.MyQueue.EnterQueue(self)
15    Passivate                          # reactivated by ticket-scan process
16    ArrivalTime = Now
17    Save Visitor Data

```

TCarGenerator

```

1     CarCapacity: TUniformDistribution
2

```

```

3 Process
4     Repeat
5         while not CarFree:                # wait until a car slot is free
6             Standby
7             newCar = new TCar
8             newCar.CC = CarCapacity.sample
9             newCar.Start

TCar
1     CC:                integer                # car capacity (number of passengers)
2     Passengers:        list of TVisitor
3     NodeLog:           list of (Node, Time)
4     Waittime:          integer
5     MaxWaitTime:       integer
6     ParkingLotQueue:   integer                # shared counter: cars queuing at
        parking entrance
7     CarsInParkingLot: integer                # shared counter: cars currently in
        parking lot
8
9 Process
10    Waittime = Now
11    CarFree = False                # mark slot as occupied
12    # wait until enough passengers have queued OR timeout
13    while MyCarQueue.Length < CC:
14        if (Now - Waittime) > MaxWaitTime:
15            break                # depart with whoever is in the queue
16        Standby
17    for i in range(MyCarQueue.Length):
18        nextVisitor = MyCarQueue.FirstOfQueue
19        nextVisitor.LeaveQueue(MyCarQueue)
20        Passengers.Add(nextVisitor)
21    CarFree = True                # signal generator: slot free again
22    Route = RoadNetwork.GetRoute(Spawnpoint)
23    CurrentNode = Route.FirstNode
24    TrafficDensity[CurrentNode] += 1
25    NodeLog.Add(CurrentNode, Now)
26    for each NextNode in Route:
27        Speed = TrafficSpeedFunction(TrafficDensity[CurrentNode])
28        Distance = RoadNetwork.GetDistance(CurrentNode, NextNode)
29        TravelTime = Distance / Speed
30        Wait TravelTime
31        TrafficDensity[CurrentNode] -= 1
32        CurrentNode = NextNode
33        TrafficDensity[CurrentNode] += 1
34        NodeLog.Add(CurrentNode, Now)
35    ParkingLotQueue += 1
36    Wait ParkingLotEntryDelay.sample * ParkingLotQueue
37    ParkingLotQueue -= 1
38    CarsInParkingLot += 1
39    NodeLog.Add(ParkingLot, Now)
40    Wait FindSpaceParkCar * CarsInParkingLot
41    NodeLog.Add(Parked, Now)
42    for each Visitor in Passengers:
43        Visitor.Reactivate
44    Save NodeLog
45    Passivate

TShuttleBus
1     BusCapacity:       integer
2     MaxWaitTime:       integer
3     Passengers:        list of TVisitor
4     NodeLog:           list of (Node, Time)
5     TripLength:        list of (Trip, Time)
6     Waittime:          integer

```

```

7      Depart:          integer
8      Arrive:          integer
9      Trip:            integer
10     BusEquivalent: integer          # PCU equivalent of the bus for traffic
                                     density
11
12 Process
13     CurrentNode = RoadNetwork.GetRoute(Station).FirstNode
14     TrafficDensity[CurrentNode] += BusEquivalent
15     NodeLog.Add(CurrentNode, Now)
16     Repeat
17         # --- boarding phase ---
18         Waittime = Now
19         while MyBusQueue.Length < BusCapacity:
20             if (Now - Waittime) >= MaxWaitTime:
21                 break                # depart even if not full
22             Standby
23         boardcount = Min(BusCapacity, MyBusQueue.Length)
24         for i in boardcount:
25             nextVisitor = MyBusQueue.FirstOfQueue
26             nextVisitor.LeaveQueue(MyBusQueue)
27             Passengers.Add(nextVisitor)
28         BoardingDone = Waittime + boardcount * BoardingTimePerPassenger
29         Wait Max(0, BoardingDone - Now)          # only wait the remaining
                                                  boarding time; wait already spent counts
30         # --- drive to festival ---
31         Depart = Now
32         Route = RoadNetwork.GetRoute(BusToFestival)
33         for each NextNode in Route:
34             Speed = TrafficSpeedFunction(TrafficDensity[CurrentNode])
35             Distance = RoadNetwork.GetDistance(CurrentNode, NextNode)
36             TravelTime = Distance / Speed
37             Wait TravelTime
38             TrafficDensity[CurrentNode] -= BusEquivalent
39             CurrentNode = NextNode
40             TrafficDensity[CurrentNode] += BusEquivalent
41             NodeLog.Add(CurrentNode, Now)
42         # --- discharge passengers ---
43         Arrive = Now
44         TripLength.Add(Trip, Arrive - Depart)
45         Trip += 1
46         Wait Passengers.Length * AlightingTimePerPassenger # alighting takes
                                                  time per passenger
47         for each Visitor in Passengers:
48             Visitor.Reactivate
49         Passengers.Clear
50         # --- drive back to station ---
51         Wait EmptyBus
52         Route = RoadNetwork.GetRoute(BusToStation)
53         for each NextNode in Route:
54             Speed = TrafficSpeedFunction(TrafficDensity[CurrentNode])
55             Distance = RoadNetwork.GetDistance(CurrentNode, NextNode)
56             TravelTime = Distance / Speed
57             Wait TravelTime
58             TrafficDensity[CurrentNode] -= BusEquivalent
59             CurrentNode = NextNode
60             TrafficDensity[CurrentNode] += BusEquivalent
61             NodeLog.Add(CurrentNode, Now)
62         ArriveAtStation

TTicketScan
1      ScanTicket: integer          # fixed time to scan one ticket
2
3 Process

```

```

4   Repeat
5       while MyQueue.Length == 0:
6           Standby                # wait (standby) until a visitor is in the
                                   queue
7           nextVisitor = MyQueue.FirstOfQueue
8           nextVisitor.LeaveQueue(MyQueue)
9           Wait ScanTicket
10          nextVisitor.Reactivate

```

Initialization

```

1   RoadNetwork.Load(GoogleMapsData)
2   TrafficDensity.Initialize(RoadNetwork)
3   TicketScan = new TTicketScan
4   TicketScan.Start
5   Shuttlebus = new TShuttleBus
6   Shuttlebus.Start
7   CarGenerator = new TCarGenerator
8   CarGenerator.Start
9   VisitorGenerator = new TVisitorGenerator
10  VisitorGenerator.Start

```

8 Data Collection and System Analysis

For the simulation model, two datasets are used: one containing the ticket scan data from the festival in 2025, and one containing all traffic data. These datasets are used for validation and calibration of the simulation model, as well as to analyze system performance. This chapter describes the datasets, explains how they are processed, cleaned and prepared as input for the simulation model.

8.1 Datasets

Ticket scan data This Excel-file was sent by the Ribs & Blues festival organization. The dataset contains historical ticket scan records from the Ribs and Blues festival in 2023, 2024, and 2025. The data records the cumulative number of visitors entering the festival site every half hour and may be influenced by external factors such as weather conditions. In addition, in 2023 and 2024, visitors were able to collect their tickets before the start of the festival, so they differ from the data for 2025.

Traffic data: This dataset contains historical traffic data for three selected locations near the festival. This set was obtained from the Nationaal Dataportaal Wegverkeer (NDW). The data consists of the number of vehicles passing each location per hour over three weekends in June. Namely, weekend during which the festival took place in 2025, and the corresponding weekdays in the two following weeks. By using the traffic data of when the festival took place, and that of the same days in the week, it can be derived how the traffic differs from regular traffic.

8.2 Processing and cleaning of data

Ticket scan data Figure 2 shows the raw data of the number of ticket scans per half hour. Figure 3 visualizes these data and shows the tickets can pattern over time. This contains the data from 2025, since this is the data closest to only entering with tickets and not on beforehand distributed tickets.

7-6-2025 tickets/half hour		8-6-2025 tickets/half hour		9-6-2025 tickets/half hour	
12:30	0	12:30	76	12:30	183
13:00	0	13:00	1070	13:00	1236
13:30	0	13:30	1069	13:30	1252
14:00	0	14:00	822	14:00	971
14:30	0	14:30	762	14:30	593
15:00	0	15:00	553	15:00	778
15:30	0	15:30	996	15:30	487
16:00	0	16:00	610	16:00	247
16:30	0	16:30	415	16:30	196
17:00	0	17:00	224	17:00	111
17:30	0	17:30	142	17:30	88
18:00	97	18:00	100	18:00	68
18:30	615	18:30	164	18:30	8
19:00	1045	19:00	148	19:00	1
19:30	830	19:30	172	19:30	0
20:00	503	20:00	170	20:00	0
20:30	652	20:30	115	20:30	0
21:00	760	21:00	100	21:00	0
21:30	311	21:30	45	21:30	0
22:00	114	22:00	17	22:00	0
22:30	60	22:30	4	22:30	0
23:00	31	23:00	0	23:00	0
23:30	6	23:30	0	23:30	0

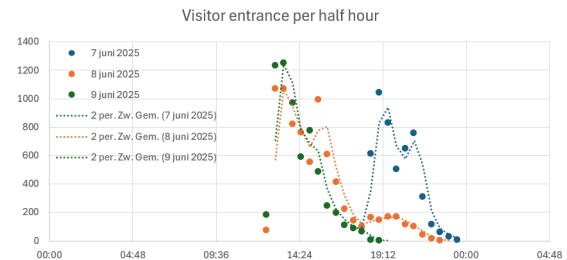


Fig. 3. Visualization tickets scanned/half hour

Fig. 2. Number of tickets scanned per half hour

Based on a first look on the visualization of the ticket scan data in Excel, the visitor arrival pattern appears to follow a gamma distribution. To further look into this, a Python script was developed. This script is shown in Appendix A. This script creates an interactive Matplotlib visualization that compares the observed festival arrival data with a theoretical gamma distribution curve. The interactive visualization includes sliders and radio buttons that allow the user to change the distribution parameters. Namely, the shape, scale, and shift, and the user can change between the three festival days. Ultimately, this allows to visually experiment and find the best-fitting model for the observed scanner data.

Traffic data The data of the traffic is an Excel sheet with four columns, one column for the dates (there are three weekends: 8-10 June, 15-16 June, 21-23 June), one column for each location (as seen in the

figure), one column for each hour on each of the days, and one column for the traffic count.

The selected measurement locations represent the three main roads around Raalte, and therefore as the access routes for car traffic entering Raalte. The three locations are defined as follows:

- Location 1: GEO2A_R_RWSTI36296 (East)
- Location 2: GEO2A_R_RWSTI362781 (North)
- Location 3: POV07_N348_Hmp_64.4_Re_HTN348 (West)

The geographical location of these three measurement points can be seen below in figure 4.



Fig. 4. Geographical location of the three measurement points

The traffic data around Raalte is first prepared by converting the date variable into a datetime format. This ensures that the dates are interpreted correctly and enables chronological operations such as sorting and grouping. Next, the observations are sorted according to location, date, and hour. This creates a chronological ordering within each location. A new variable representing “weekend” periods is then introduced. Instead of using conventional calendar weekends, three unique dates are grouped into a single weekend identifier. This way, the data are partitioned into sequential three-day periods that can be analyzed and compared. Finally, the dataset is aggregated by weekend, location, and hour. For each combination of these variables, the mean number of vehicles is calculated. The resulting dataset therefore provides the average hourly traffic volume for each location during each defined weekend period.

Using a Python code, the data was then plotted in a graph. For this visualization the averages of the three-day weekend per location were used. An important note is that the festival took place in the first weekend. This visualization of the traffic data can be seen below in figure 5.

This is also done for the second and third weekend.

8.3 Preparation for data input

8.4 Ticket scan data

The processed ticket scan data is used as input for the simulation model, and can be used to validate whether the simulated arrival pattern follows the observed visitor arrival distribution. Figure 6 shows a fitted gamma distribution for the ticket scans for the first festival day.

The observed ticket scan data contains some bias. The festival organization who sent us the ticket scan data mentioned that the second peak in the arrival pattern was unusual and unexpected, and was likely caused by poor weather conditions. For this reason, the gamma distribution was fitted in such a way that it fits the overall arrival pattern, without adjusting to this irregular second peak in the observed data.

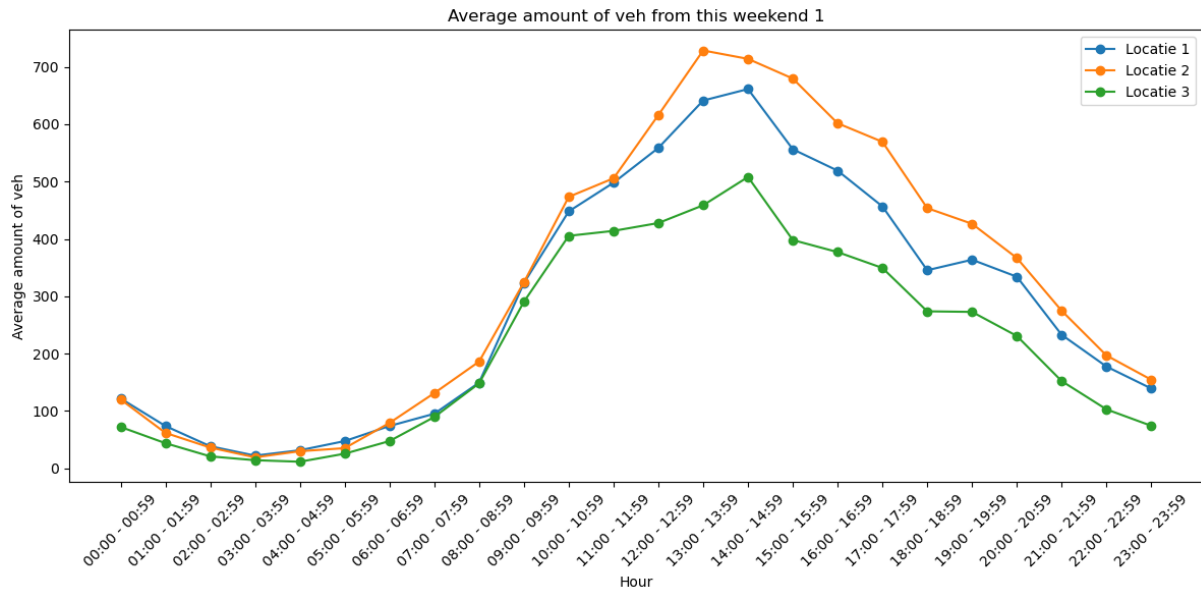


Fig. 5. Average traffic count over first weekend, at the three locations in Raalte

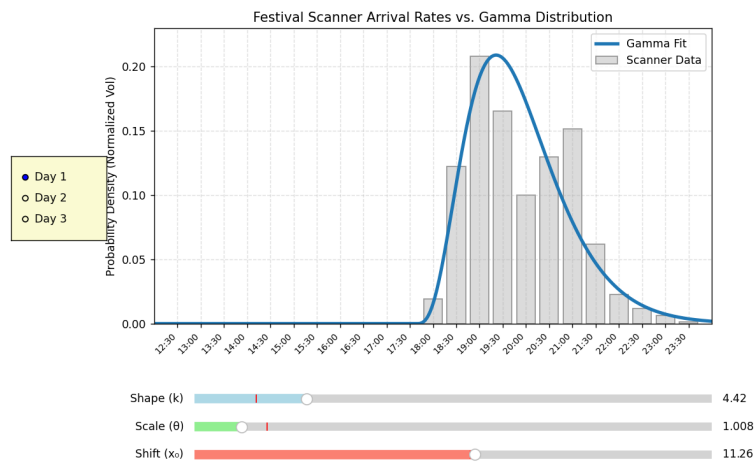


Fig. 6. Gamma distribution on festival data

For each of the three different festival days, different distribution parameters were obtained. These parameter values are presented in table 1.

	Day 1	Day 2	Day 3
Shape (K)	4.4	1.8	2.2
Scale (θ)	1.00	2.9	1.7
Shift (X_0)	11.3	0.48	0.48

Table 1. Scanner arrival rate gamma distribution values

Traffic data: After preparing the data, the hourly traffic counts show a characteristic shape in which the number of vehicles is relatively low during the night and early morning hours of the day, gradually

increases towards a peak during busy traffic periods, and then decreases again later in the day. This pattern resembles the shape of a Gamma distribution. Therefore, a gamma distribution function is suitable for modeling the temporal behavior of traffic flow in the PDL model.

The prepared dataset was first grouped into “weekends,” where every three consecutive days were treated as one weekend period. Subsequently, the average number of vehicles per hour was calculated for each weekend and each location. Taking the average reduces random fluctuations in the measurements and creates a smoother traffic profile that better represents the general traffic behavior throughout the day.

The averaged traffic count data was then stored in a dictionary structure. Separate datasets were created for each location, as well as an additional dataset containing the average of all locations combined. This structure allows the different traffic patterns to be selected and visualized easily within the program.

An interactive visualization was created using Matplotlib widgets. The normalized traffic count data are displayed as bar charts, while a Gamma distribution is plotted as a continuous curve on top of the bars. The radio buttons allowed us to select different weekends (we can choose between each weekend and location, or average per weekend). Sliders make it possible to adjust the parameters of the Gamma distribution interactively.

The parameter k controls the shape of the Gamma distribution and determines how sharp or skewed the curve appears. The parameter θ controls the spread of the curve and determines how wide the distribution becomes. In addition, a shift parameter moves the Gamma distribution horizontally along the time axis so that the peak of the curve can align with the observed traffic peak in the data. By adjusting these sliders, the Gamma distribution can be visually fitted to the observed traffic count data.

Using this fitting process suitable values for the Gamma distribution parameters can be determined. The fitted Gamma distributions can subsequently be used as input for the PDL model to represent the temporal distribution of traffic flow. The figure below shows how the interactive plot works.

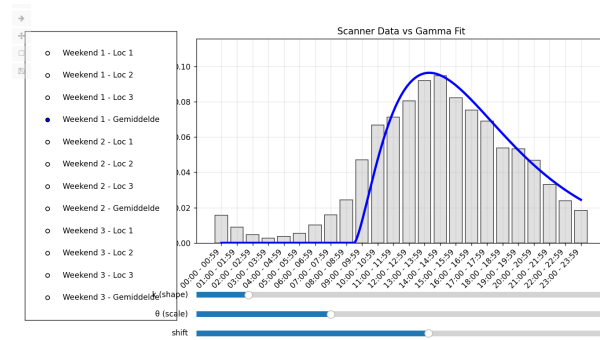


Fig. 7. Enter Caption

For the weekend in which the festival took place, the best fitting distribution parameters were obtained. These parameter values are presented below in table 2

Shape (K)	2.5
Scale (θ)	3.2
Shift (X_0)	8.5

Table 2. Scanner arrival rate gamma distribution values

8.5 Insights about data

As previously explained, the patterns in both the ticket scan data and the traffic data are best represented by a gamma distribution. A gamma distribution is described and defined by three parameters: the shape, the scale, and the shift parameter. The shape parameter gives form to the curve and affects how sharply the arrival peak develops. The scale parameter controls how the distribution evolves over time, and the shift parameter allows the start of the distribution to be adjusted as wished.

Ticket scan data The tickets scan data shows how the people behave when entering the festival. The artists were starting on the first day at 19:30. The majority of the guest were entering around 19:00, to see the artists. On the second and third day, the same can be said. The festival started at 13:30 and most people were entering around 13:00. So it is in the behavior of the people that most of them enter the festival terrain half an hour before the start. At its core, the Gamma distribution predicts the time you will wait until a specific number of events occur. In the context of a festival, an "event" is someone successfully passing through the security or ticket check.

Traffic data The traffic data reveals the daily travel patterns on the main access roads around Raalte. Unlike the ticket scan data, which is heavily concentrated around specific festival start times, the traffic data captures a broader daily rhythm. Typically, traffic volume is minimal during the night, begins to rise in the morning (as reflected by the shift parameter X_0 of 8.5 in the first weekend, indicating an initial increase in traffic around 08:30 AM), reaches a peak during the day, and tapers off in the evening.

In the context of traffic flow, the Gamma distribution models how vehicle arrivals are distributed throughout the day. The shape (K) and scale (θ) parameters describe the concentration of the traffic. A higher shape parameter coupled with a specific scale indicates a sharp, condensed peak in traffic (such as rush hour), whereas a different configuration might describe a wide, steady flow of vehicles over several hours.

9 Full simulation

10 Simulation Implementation

Describe your discrete event simulation model here. Include:

- *The software or programming language used.*
- *The conceptual model and core logic (entities, resources, events).*
- *Assumptions and simplifications made during implementation.*

11 Experimental Evaluation

11.1 Experimental Setup

Explain the different conditions, scenarios, and parameters you tested. What were your independent and dependent variables? How many replications did you run?

For the experimental evaluation, certain tests are performed. The goal of this experimental plan is to evaluate the impact of visitors of festivals on the environment and look for ways to reduce the CO₂-emission. Tests that has been performed are:

- Train ticket discount: Ranging from 0% to 100%
- Sustainability fee: Ranging from 0% to 40%
- Shuttle bus frequency: Ranging from 3 to 6 departures per hour
- Parking costs: Ranging from 0% to 50% added to the base ticket price.

The dependent variables (KPIs) measured across these scenarios included the resulting modal split, total CO₂ emissions per passenger and peak queue lengths at the ticket scan and shuttle stops.

The baseline output is the current situation.

11.2 Performance Analysis

Present the baseline results of how the system performs. Use tables and figures to illustrate key performance indicators (KPIs) like wait times, throughput, or resource utilization.

12 Impact of Proposed Innovation

Describe the proposed innovation you introduced to the model. Compare the simulation results of the baseline system against the results of the system with the proposed innovation. Did performance improve? By how much?

13 Conclusions and Recommendations

Summarize all your findings based on the data. Provide clear, actionable recommendations for real-world system improvements based on your simulation's outcomes.

14 Ai Statement

For this report, Generative AI has been used to:

1. Improve the grammar, style, layout, and/or spelling of the text

During the preparation of this work, we used the Overleaf built-in Writefull language support to correct grammar and spelling mistakes and improve the overall readability. In all cases, we have reviewed and corrected work and remain fully responsible for the content of the report.

References

1. Bottrill, C., Papageorgiou, S., & Jones, M. (2009). Jam packed: Audience travel emissions from festivals. Julie's Bicycle. <https://www.greeneventbook.com/wp-content/uploads/2024/01/jb-TravelReport-sm-may09.pdf>
2. Vereniging Nederlandse Poppodia en -Festivals. (2026, 10 februari). Slimmer en duurzamer naar het festival: lessen uit vier mobiliteitspilots. <https://vnpf.nl/nieuws/slimmer-en-duurzamer-naar-het-festival-lessen-uit-vier-mobiliteitspilots>
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Appendix A

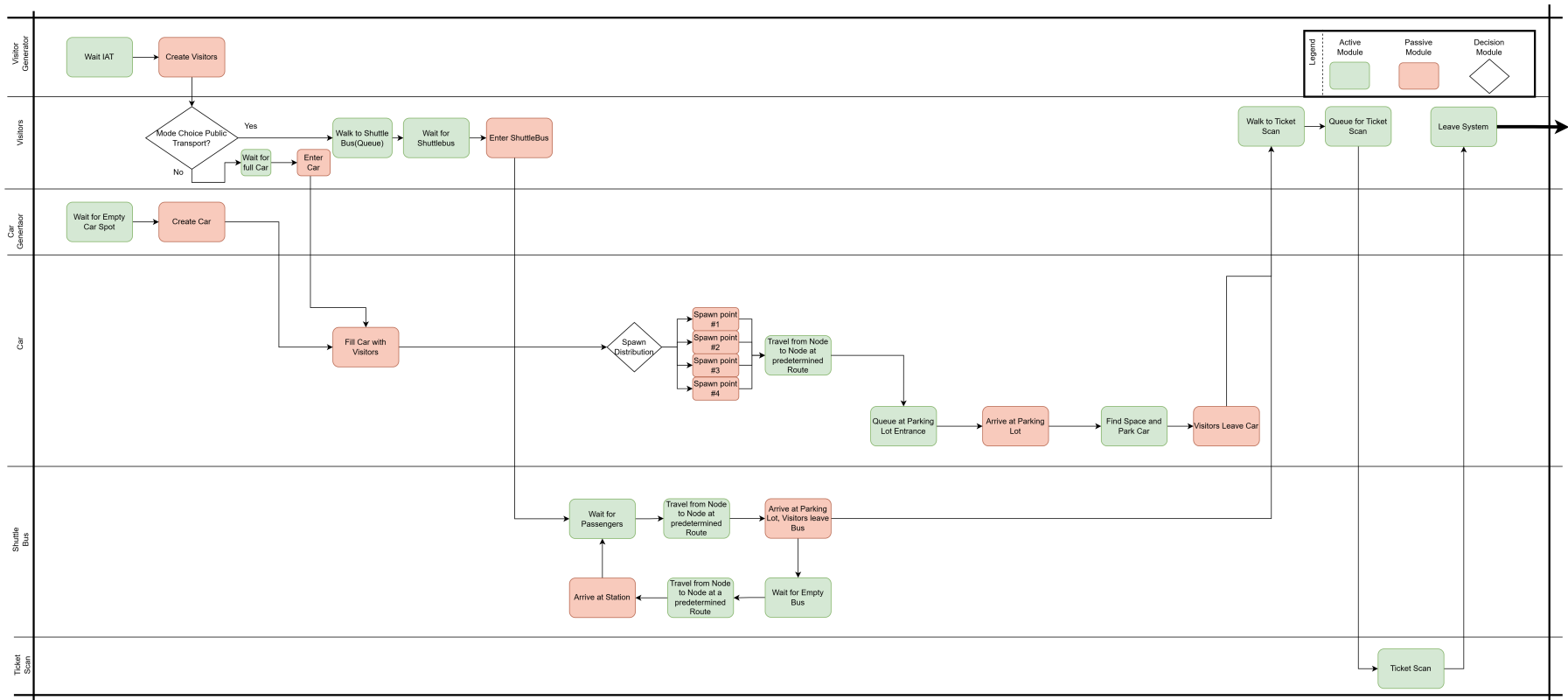


Fig. 8. Swim-lane Diagram

A Python ticket cleaning

[illegible]

```

# [left , bottom , width , height]
ax_k = plt.axes([0.25, 0.2, 0.65, 0.03])
k_slider = Slider(ax=ax_k, label='Shape (k)', valmin=0.1, valmax=20.0, valinit=init_k, c

ax_theta = plt.axes([0.25, 0.15, 0.65, 0.03])
theta_slider = Slider(ax=ax_theta, label='Scale (theta)', valmin=0.1, valmax=10.0, valin

ax_shift = plt.axes([0.25, 0.1, 0.65, 0.03])
shift_slider = Slider(ax=ax_shift, label='Shift (x0)', valmin=-5.0, valmax=25.0, valinit

# 5. Create Radio Buttons for Day Selection
ax_radio = plt.axes([0.02, 0.5, 0.12, 0.15], facecolor='lightgoldenrodyellow')
radio = RadioButtons(ax_radio, ('Day 1', 'Day 2', 'Day 3'))

# 6. Define the Update Function
def update(val):
    # Read values
    k = k_slider.val
    theta = theta_slider.val
    shift = shift_slider.val
    day = radio.value_selected

    # 1. Update the Bar Chart
    new_norm_data = normalize_data(data_dict[day])
    for bar, h in zip(bars, new_norm_data):
        bar.set_height(h)

    # 2. Update the Line Chart
    y_new = gamma.pdf(x_line, a=k, loc=shift, scale=theta)
    line.set_ydata(y_new)

    # 3. Dynamically adjust Y-axis limit so nothing gets cut off
    max_data = max(new_norm_data)
    max_curve = max(y_new)
    ax.set_ylim(0, max(max_data, max_curve) * 1.1)

    # Redraw
    fig.canvas.draw_idle()

# Link everything to the update function
k_slider.on_changed(update)
theta_slider.on_changed(update)
shift_slider.on_changed(update)
radio.on_clicked(update)

# Show the interactive plot
plt.show()

```